How Much Weather Is Enough for Predicting Renewable Grid Share?

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Research Question

1. How much accuracy ERA5 wind, radiation, and temperature add beyond calendar features;
2. Whether this gain is predictable from a grid's wind-versus-solar mix;
3. How far resolution can be coarsened, and whether weighting cells toward mapped generation facilities matters more than grid fineness.

Data

- We analyzed 19 European countries from 2022 to 2026.
- Used 36,000–38,000 hourly Energy-Charts observations per country.
- Combined electricity data with ERA5 weather weighted by wind and solar farm locations and capacities from Global Energy Monitor.

Methodology

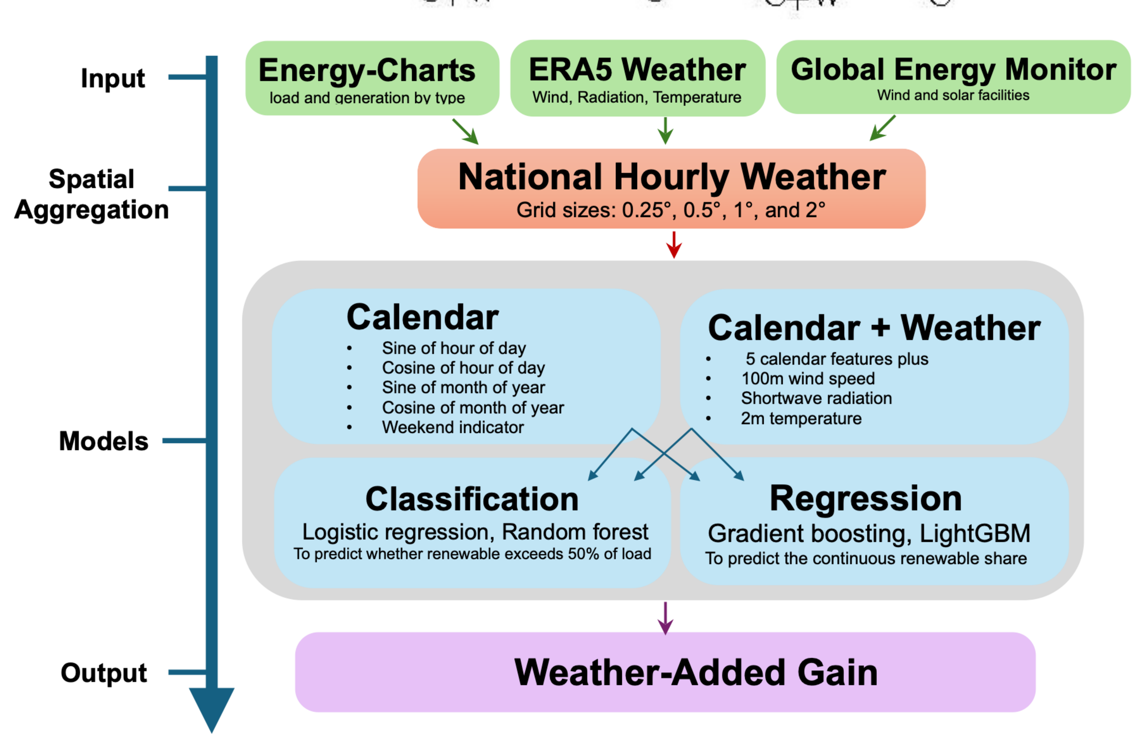
- Two feature sets:
  - ✓ Calendar (C): five calendar features
  - ✓ Calendar + Weather (C+W): five calendar features + 3 weather features
- Two predictions:
  - ✓ Classification: (a) To predict whether renewable share R_t exceeds 50% of load; (b) Area Under the Curve (AUC) metric.
  - ✓ Regression: (a) To predict the continuous renewable share R_t; (b) coefficient of determination R² and Mean Absolute Error (MAE) metrics.
- Weather-added gain:
  - ✓ To analyze whether G increases with the wind dominance using the Pearson correlation r

$$R_t = \frac{\text{renewable generation at hour } t}{\text{electrical load at hour } t} \times 100.$$

$$r = \text{Pearson-Corr}(D, G)$$

$$D = \left(\frac{\sum_t \text{wind generation}_t}{\sum_t \text{load}_t} - \frac{\sum_t \text{solar generation}_t}{\sum_t \text{load}_t}\right) \times 100$$

$$G = \text{AUC}_{\text{C+W}} - \text{AUC}_{\text{C}} \text{ or } R^2_{\text{C+W}} - R^2_{\text{C}}$$



References:

AI disclosure: OpenAI Codex assisted with code, figures, and editing. All results and conclusions were verified by the authors.

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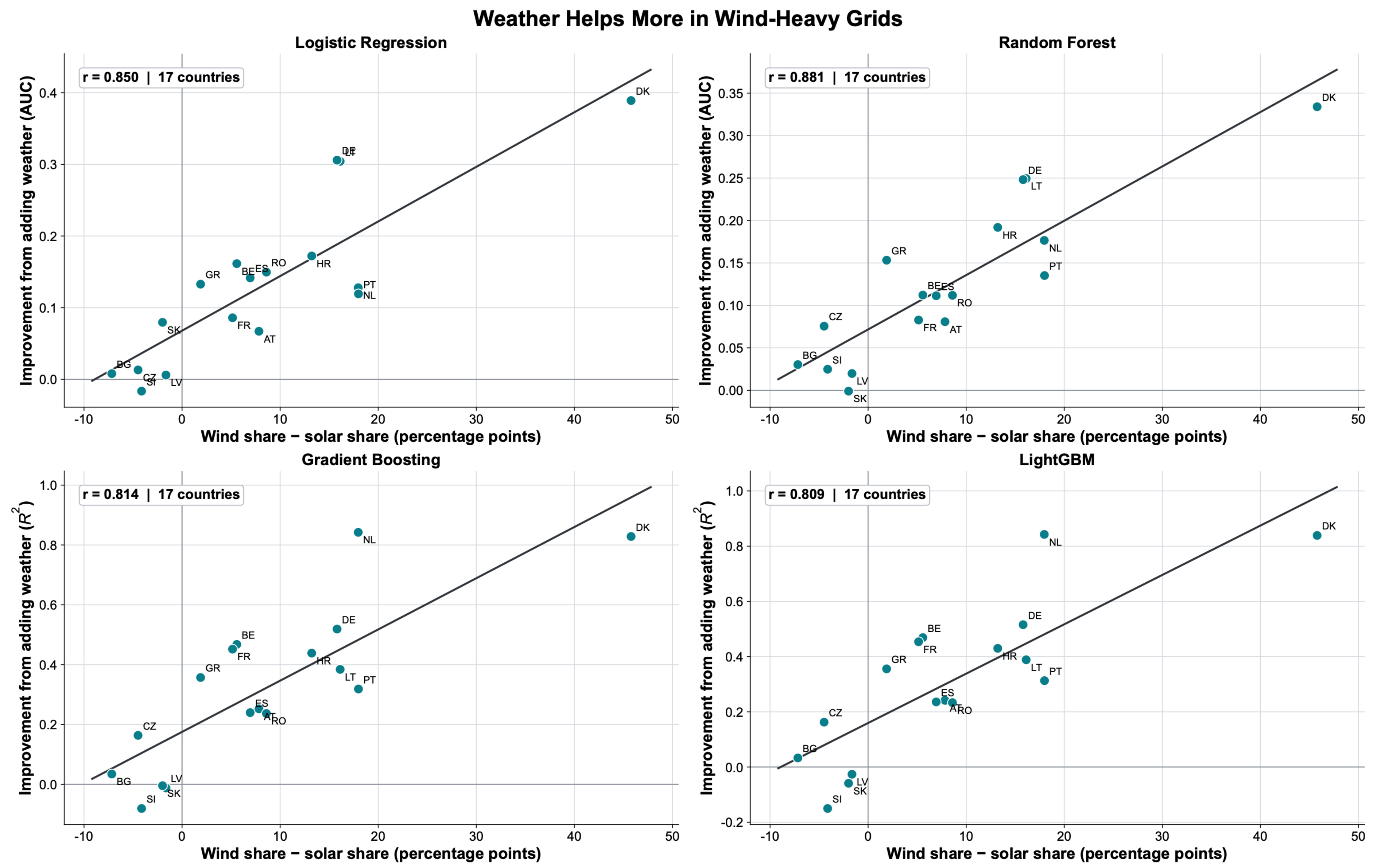
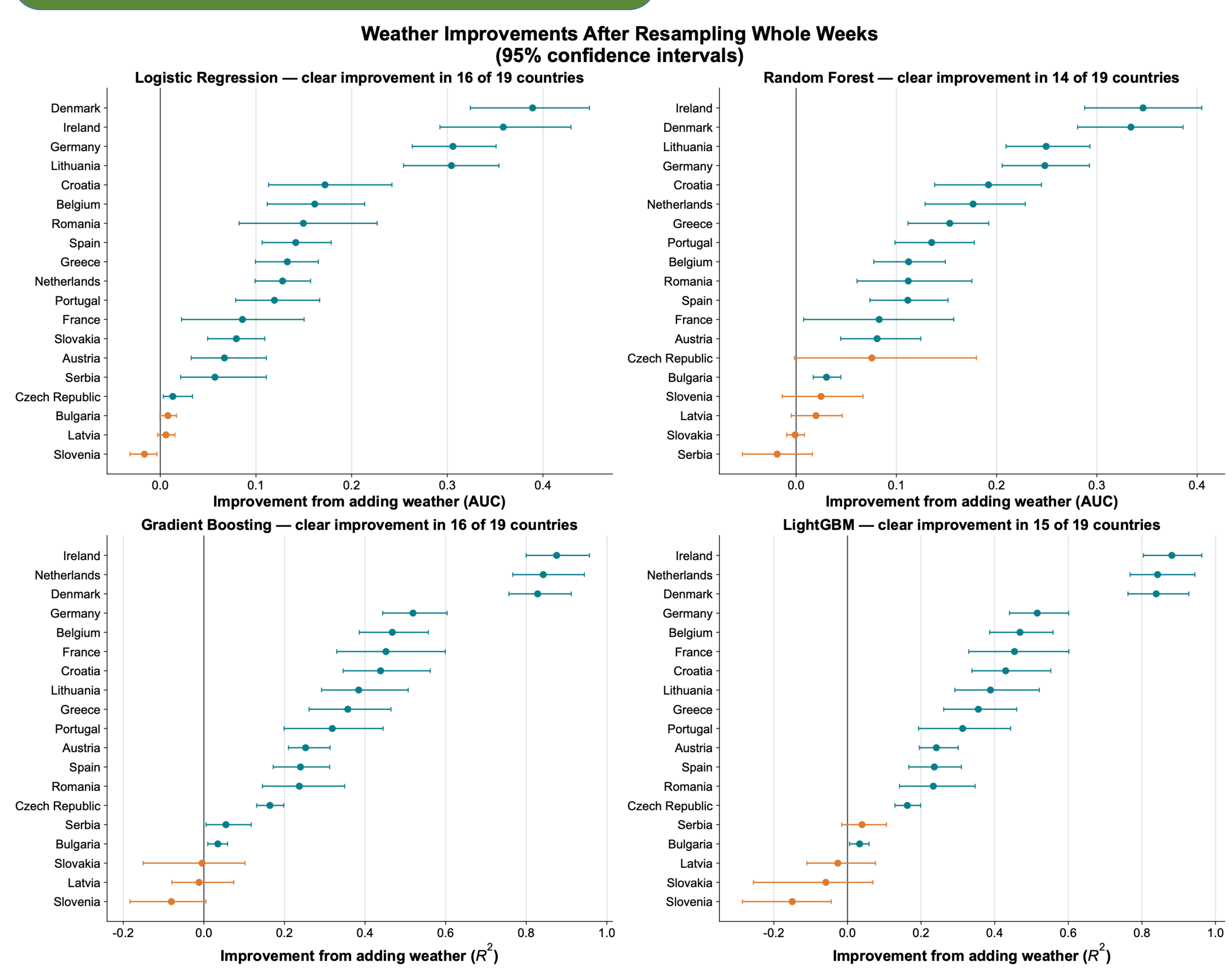
Wind-Dominant Grids Benefit from Weather



- Weather-added gain increased with wind dominance for all four models.
- Correlations ranged from $r = 0.809$ to $r = 0.881$
- The relationship used 17 grids with complete wind and solar data.

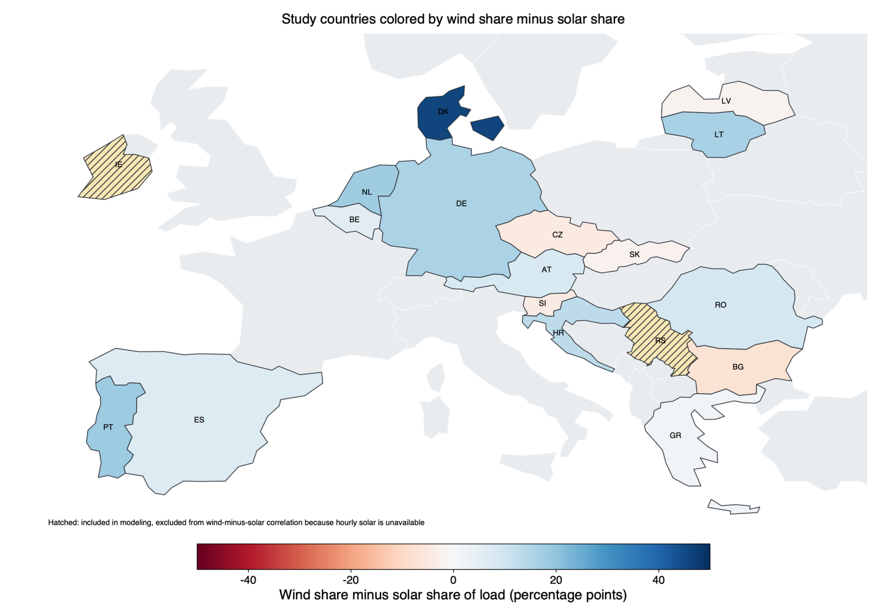
Weather-Added Gains Are Robust



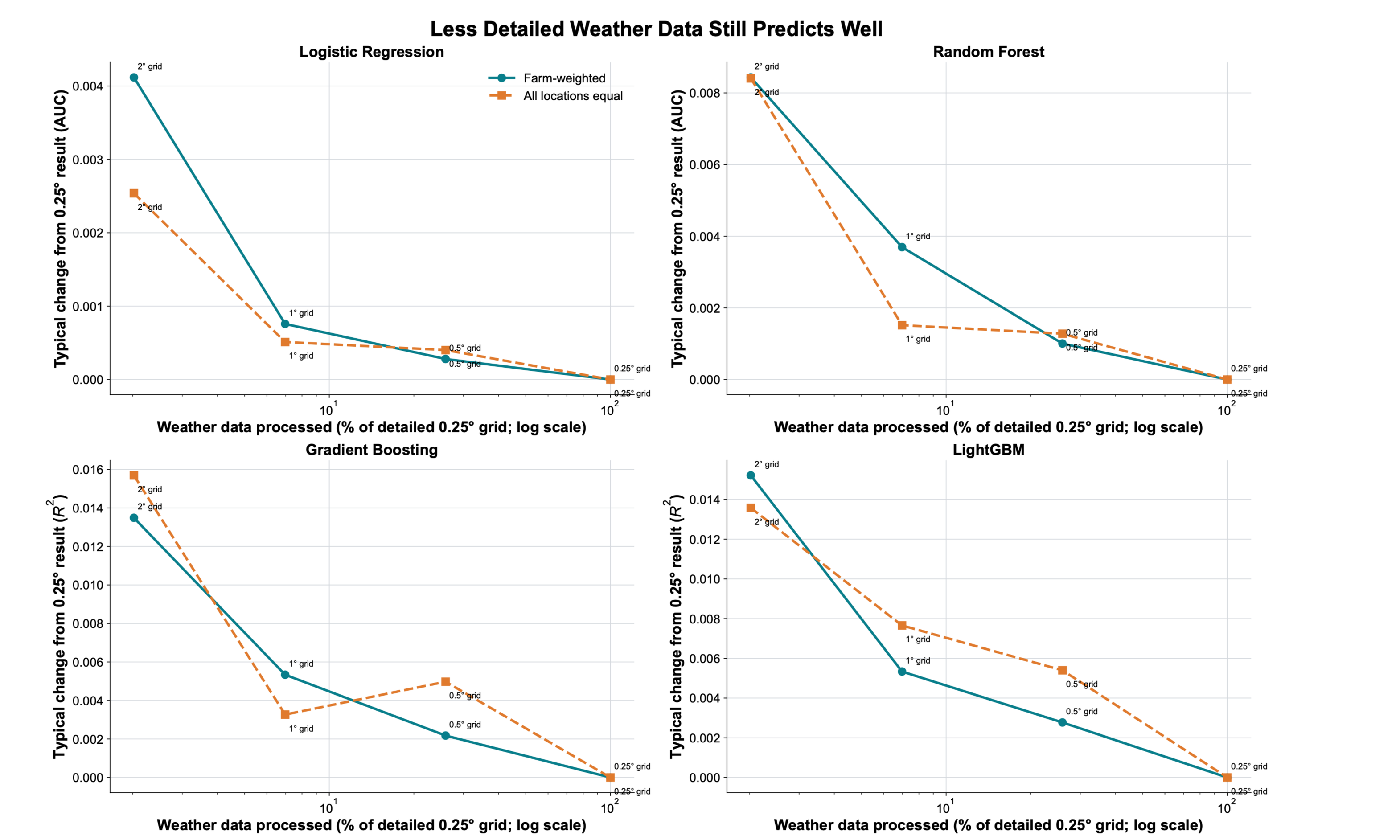
- Adding weather clearly reduced prediction errors in most countries, depending on the model.
- Wind-heavy countries generally showed the largest and most consistent improvements.

Wind- vs Solar-Dominant Systems

- Study countries colored by wind minus solar share of load.
- Wind-dominant systems appear dark blue; Solar-dominant systems appear in warm tones.
- Ireland and Serbia lack solar observations and are marked unavailable.

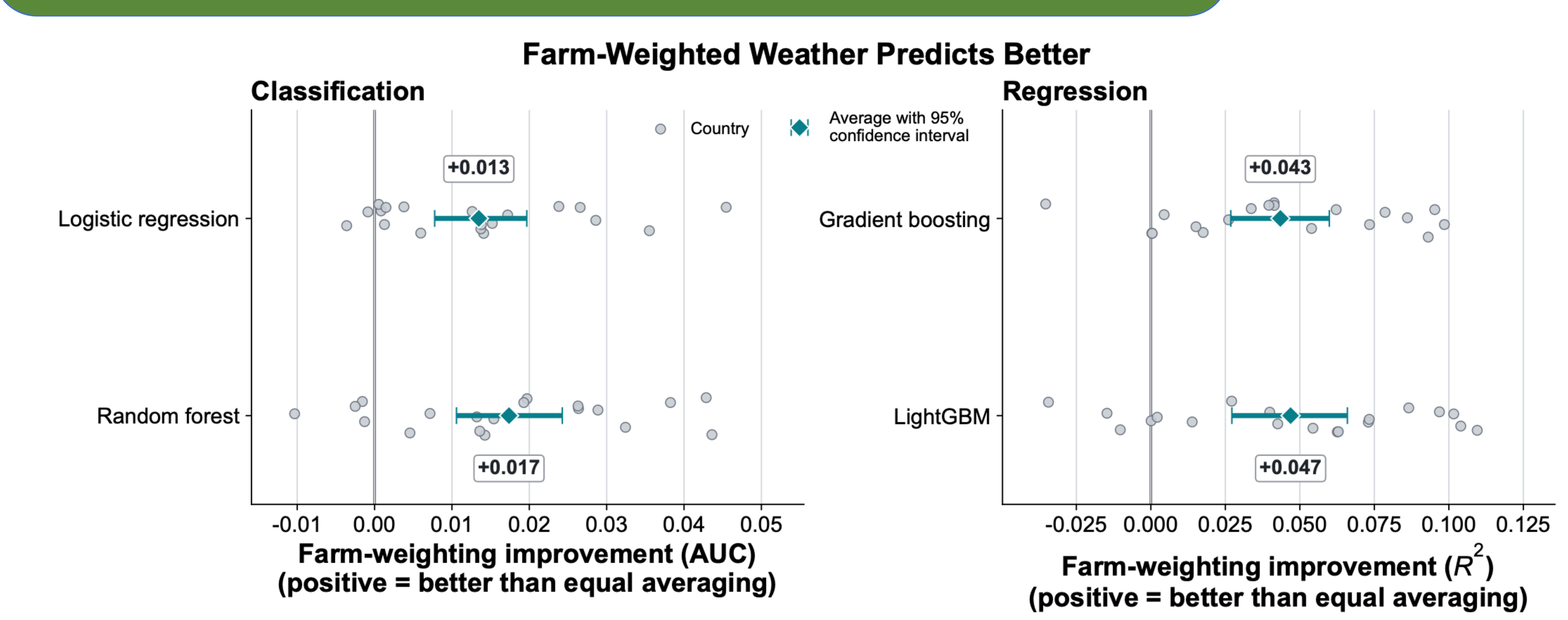


Coarser Weather Grids Preserve Accuracy



Moderate coarsening maintained nearly the same prediction quality while reducing weather data processed from 100% at 0.25° to 25.9% at 0.5° and 7.0% at 1°.

Power-Plant-Weighted Weather Predicts Better



- Weighting weather toward wind and solar farms usually outperformed treating all locations equally.
- Average improvements were 0.013–0.017 AUC and 0.043–0.047 R^2, although some countries did not improve.

Conclusions

- Weather improved renewable-share predictions in most countries.
- Wind-heavy grids benefited most from weather data.
- Coarse grids are enough: 0.5°–1° grids preserve nearly all predictive value